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Using the handlebar extension prototype, our team employed the help of Vanderbilt engineers Ms. Piper Cannon, a grad student at VISE, and Mr. Jared Lawson, a current PHD candidate at Vanderbilt, in order to help us troubleshoot problems. However simple, our design allowed us to fulfill the safety and manufacturability principles stated by Cambridge Design Technology to be two of four main engineering design principles one must take into consideration when designing a product. Despite the success of our design in two aspects, flaws were found in the functionality and the reliability of the design. During the classroom Designathon, Vanderbilt engineers used their expertise to help us find solutions to our design's pitfalls. Starting with Engineer Ms. Cannon: She brought up how grips were a good idea and how a pool noodle or caulk could be good for it She said the duct tape is ugly, but it is good if we wanted to change the length of the velcro She also stated that we should replace the 2nd gripping pipe to resemble the first gripping piece better, so it is more even and surrounds the cart handle better (cover more of the handle) Once Ms. Canin concluded her critiques, we got to work putting her words into action. During this process, Engineer Mr. Lawson stopped by to add additional commentary about the design. Lawson explained that instead of trying to re-fasten the gripping pipe to cart handle using two screws, we should let the pipe be able to shift around. The critiques given during this Designathon were promptly acknowledged and put into action. In the later days, our team addressed a need for grips in the gripping pipes and used plastic tubing to achieve this result. Using Mr. Lawson's advice, our team chose not to screw in the second screw into one of the posts in fear of splintering and for mobility. Through these ideas, our team inches closer to achieving functionality and reliability. The proposed solution for Mr. Kiser's handlebar extension is thoroughly grounded in STEM principles, particularly in the domains of safety and

manufacturability. Our team adhered to key design principles outlined by Cambridge Design Technology, which emphasizes safety and manufacturability as two of the four fundamental engineering design criteria that must be taken into account when creating a product. By focusing on these principles, we ensured that our handlebar extension prototype would not only meet the functional needs of the user but also be safe and feasible to produce at scale. However, while the design successfully addressed safety and manufacturability, our initial prototype revealed some functional and reliability issues. These shortcomings were identified during our Designathon, where we were fortunate to have the expertise of Vanderbilt engineers, Ms. Cannon and Mr. Lawson, who helped us identify and troubleshoot key problems with the design. Their input, based on years of engineering experience, was instrumental in guiding our team toward a more refined solution. Engineer Ms. Cannon offered a series of critiques and suggestions that proved invaluable. She suggested that grips would enhance the comfort of the handle, and proposed using materials like pool noodles or caulk to create a more ergonomic grip. Additionally, she pointed out that while duct tape was functional, it detracted from the aesthetic and should only be used for temporary adjustments, such as when experimenting with the length of Velcro attachments. Ms. Cannon also recommended that we modify the second gripping pipe to better resemble the first, ensuring it fit the cart handle more evenly and provided a greater surface area for contact. These suggestions were immediately recognized by our team as important steps to improve both the design's usability and visual appeal. Once Ms. Cannon had shared her input, we quickly got to work incorporating her suggestions. During this process, Mr. Lawson provided further insights that were crucial to the functionality of the handlebar extension. He also recommended that instead of securing the second gripping pipe to the cart handle with two screws, we should allow the pipe to shift freely. This advice was given with the goal of avoiding

unnecessary rigidity in the design and ensuring that the handle could adapt to slight variations in the cart's dimensions or user preference. Mr. Lawson also noted that by not attaching the second screw, we would reduce the risk of splintering, and the handle would be more mobile, while also being adaptable. Acting quickly, our team turned these expert reviews from theory into action; they were applied immediately in our design process. Following the feedback, we made several key improvements. We used plastic tubing to provide the necessary grip on the pipes, addressing both comfort and safety concerns. We also heeded Mr. Lawson's advice by choosing not to install the second screw, thus enhancing the mobility of the handlebar extension while minimizing the risk of material damage. By incorporating the guidance provided by two qualified engineers, our team was able to refine our design, moving closer to a product that is both functional and reliable. The design now addresses the user's needs for adjustability, comfort, and aesthetics, while ensuring that the handlebar extension remains durable and manufacturable. These revisions also demonstrate an application of STEM principles, such as material selection, structural integrity, and user-centered design, which collectively helped to improve the overall quality and performance of the handlebar extension prototype.



Sources:

[Engineering Design Principles & Methodology - Cambridge DT](#)